

Assignment-1: Quantitative Financial Market Analysis

Course: Computational Financial Modelling Lab

Root Directory Files

1-Q1.txt

Answer and explanation for **Question 1**, focusing on introductory concepts and background analysis.

2-Q2.txt

Written analysis for **Question 2**, covering index comparison and sector-based reasoning.

3-Q3.txt

Written observations and interpretations related to **Question 3**, including global slowdown analysis.

4-IBEX35Constituents-10-25.xlsx

Compiled Excel file containing **IBEX-35 constituent history (2010-2025)** used for verification and manual inspection.

5-Q5.py

Main Python script for **Question 5**:

- Survivor detection (by years or date range)
- Canonical name handling
- Sector mapping
- Restricted price database creation
- Sector-wise survivor analysis

6-Q6.py

Python script for **Question 6**:

- Corporate action tracking
- Extraction of stock splits for 15-Year Legends
- Filtering and interpretation of true vs pseudo splits

Buildmaster.py

Utility script to build the **master price database** by combining individual stock CSV files.

SplitsDivCombo.csv

Combined CSV file summarizing **corporate action events** (splits/dividend-related data).

SplitsDividend.xlsx

Excel version of corporate action data for easier presentation and reporting.

Report.pdf

Report explaining all the components and work of the assignment.

Struct.pdf

This file addressing whole folder struct.

Help/

Assignment-1.pdf

Original assignment problem statement provided by the instructor.

Download.py

Helper script used during early stages to **download index and stock-level data**.

Guide.txt

Notes and guidance used during development (data sources, assumptions, workflow).

IBEX35/

Index.csv

Daily **IBEX-35 index-level price data**.

MasterPriceDB.csv

Consolidated **master price database** containing daily prices of all downloaded IBEX-35 stocks.

15-Year-Legends.xlsx

Final list of **companies that survived continuously in IBEX-35 over the last 15 years**, along with sectors.

IBEX35/constituents/

Contains all **constituent-related metadata**.

CompaniesList.csv

Raw list of company names extracted from multiple sources.

company_canonical_map.csv

Mapping file to normalize company names:

- RawName → CanonicalName
- BaseTicker
- Tradable flag

Used throughout the project to avoid duplication and naming inconsistencies.

ibex_YYYY_Jun.csv / ibex_YYYY_Dec.csv

Biannual IBEX-35 constituent snapshots (June & December) from **2010 to 2025**.

Used for survivor analysis.

master_company_list.csv

Union of all companies that ever appeared in IBEX-35 during the study period.

SectorMap.csv

Mapping of **CanonicalName** → **Sector**, used for sector-wise survivor analysis.

IBEX35/prices/

Contains **individual daily price CSV files** for each IBEX-35 stock downloaded via Yahoo Finance.

Example:

- `ACS.MC.csv`
- `BBVA.MC.csv`
- `ITX.MC.csv`

Each file contains:

`Date, Open, High, Low, Close, Adj Close, Volume`

IBEX35/results/

Contains **restricted price databases** generated for survivor sets.

`2010_6_to_2025_12/`

Daily price data (CSV) for each **15-Year Legend** stock, restricted to the selected date range.
Used for focused long-term analysis.

Indices/

Daily price data for **index-level comparison**:

- `NIFTY50.csv`
- `SENSEX.csv`
- `IBEX35.csv`
- `OMXS30.csv`
- `ATX.csv`

Used for normalization, correlation, and global slowdown analysis.

Q3-Plots/

Contains plots generated for **Question 3**:

- Normalized index comparison
- Drawdowns
- Relative performance
- Dashboard-style visualizations

Files:

- `Figure_1.png ... Figure_5.png`
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Summary

This project directory is organized to clearly separate:

- **Raw data**
- **Processed data**
- **Analysis scripts**
- **Results and plots**